LEGAL YEARS SYSTEM DEVELOPMENT

Phase 6: Testing Phase Document

System Testing & Quality Assurance Evaluation Report

Core Objective: To ensure that all functional modules, security protocols, performance benchmarks, and legal compliance workflows of the "Legal Years" application operate seamlessly and bug-free prior to deployment.

1.1 Executive Introduction & Objectives

Following the completion of system architecture, database modeling, user interface design, and core development in earlier phases (Phases 1 to 5), Phase 6 focuses on rigorous Quality Assurance (QA). This phase evaluates software reliability, data security, and operational readiness.

Functional Accuracy

Verifying that legal case management, client billing calculations, document search indexing, and automated court date tracking operate without flaws.

Data Privacy & Security

Ensuring attorney-client privilege protection, data encryption at rest and in transit, and strict role-based authorization rules.

1.2 Scope of Testing

- **In-Scope:** Case Document Vault, Billing & Invoicing Module, Legal Analytics Dashboard, Role-Based Access Control (RBAC), and Automated Notification Engine.
- **Out-of-Scope:** Third-Party External APIs (e.g., external government court server hardware downtime).

100%	0	< 1.2s
TEST EXECUTION RATE	CRITICAL SECURITY FLAWS	AVERAGE RESPONSE TIME

2.1 Testing Methodologies

To guarantee complete coverage, multi-tiered software testing methodologies were implemented throughout the development life cycle.

1. Unit Testing

Executed at the component level by developers to validate individual functions, algorithms (e.g., legal deadline calculator), and data input rules.

2. Integration Testing

Assessed the interaction between distinct modules, ensuring seamless data flow between the Case Management Engine and Document Storage Vault.

3. System Testing

Conducted end-to-end evaluation of the complete integrated application against non-functional and functional business requirements.

4. User Acceptance Testing (UAT)

Validated by legal domain experts, practicing advocates, and administrative staff to ensure alignment with real-world legal workflows.

2.2 Testing Stack & Environment Setup

Category	Tool / Framework	Target Testing Area
Automated Testing	Selenium WebDriver / Playwright	End-to-End User Interface & Workflow Automation
API Testing	Postman / REST Assured	RESTful API Endpoints & Microservice Integration
Performance Testing	Apache JMeter	Load Handling, Concurrent Stress, & Throughput
Security Testing	OWASP ZAP / Burp Suite	Vulnerability Scanning & Penetration Testing

3.1 Security Audit & Legal Compliance

Due to the sensitive nature of legal documents, confidentiality, integrity, and availability (CIA Triad) standards were strictly enforced and tested.

AES-256 Bit Encryption

Verified that client records, legal petitions, and evidence attachments are encrypted prior to database storage.

Role-Based Access Control (RBAC)

Tested boundary permissions ensuring junior associates, senior partners, and clients access only authorized resources.

3.2 Security Penetration Test Checklist

Security Parameter	Evaluation Description	Status
SQL Injection (SQLi) Prevention	Parameterized queries and ORM protection against malicious payload injection	**PASSED**
Cross-Site Scripting (XSS)	Input sanitization and output encoding across all input forms	**PASSED**
Multi-Factor Authentication (MFA)	Time-based OTP verification for administrative and advocate accounts	**PASSED**
Data Loss Prevention (DLP)	Restricted export permissions for confidential case files	**PASSED**
Session Timeout Management	Automatic user invalidation after 15 minutes of inactivity	**PASSED**

> **Security Audit Conclusion:** The platform successfully passed all vulnerability assessments with zero critical or high-severity vulnerabilities detected.

4.1 Defect Management Process

During the testing phase, bugs were systematically reported, categorized, prioritized, and resolved according to standard defect life cycle protocols.

Defect Severity Classification

- **Critical:** System crash, database corruption, or security breach. Requires immediate hotline fix.
- **High:** Key functionality failure without an immediate workaround (e.g., inability to upload case documents).
- **Medium:** Minor operational flaw or UI glitch where a functional workaround exists.
- **Low:** Minor cosmetic adjustments, alignment issues, or typographical fixes.

4.2 Defect Resolution Metrics

Severity Level	Reported	Resolved	Open / Pending
Critical	12	12	**0**
High	28	28	**0**
Medium	45	43	**2 (Minor)**
Low	30	28	**2 (Cosmetic)**

Defect Life Cycle Workflow

Defect Logged → Assigned to Dev → Patch Applied → Re-tested in QA → Verified → Closed

5.1 Test Execution Matrix (Sample Test Cases)

Below is a summary of primary test execution cases conducted across key system components:

Test ID	Module	Test Scenario	Expected Outcome	Result
TC_LY_001	User Auth	Login with valid advocate credentials & MFA	Redirect to Dashboard with active session	**PASS**
TC_LY_002	Case Reg	Register new case with valid client metadata	System generates unique Case File ID	**PASS**
TC_LY_003	Doc Vault	Upload encrypted PDF attachment (<25MB)	File uploaded and checksum verified	**PASS**
TC_LY_004	Billing Engine	Generate client invoice with legal fees & tax	Total calculated accurately with PDF receipt	**PASS**
TC_LY_005	Notification	Trigger automated hearing date reminder	SMS & Email delivered within 3 seconds	**PASS**
TC_LY_006	Legal Search	Search acts by legal section (e.g., IPC 302)	Relevant case law records returned instantly	**PASS**
TC_LY_007	Data Backup	Trigger scheduled automated cloud backup	Snapshot stored securely in cloud storage	**PASS**

> **Summary:** All critical test paths passed validation tests with 100% execution accuracy under simulated production loads.

6.1 Exit Criteria & Deployment Readiness

The Quality Assurance phase has met all predefined exit benchmarks. The system is certified stable and ready for final production deployment.

Exit Criteria Accomplished

- 100% planned functional test cases executed.
- Zero Critical and High defects outstanding.
- User Acceptance Testing (UAT) sign-off obtained.
- Security and performance audits completed.

Deployment Roadmap

- Production Environment Setup & Staging.
- User Onboarding & Administrative Training.
- Post-Launch Hypercare Support & Monitoring.

6.2 Project Sign-Off Approval

The undersigned project stakeholders confirm that Phase 6 (Testing Phase) has been completed satisfactorily:

Role	Name / Evaluator	Approval Status	Date
Lead QA Engineer	QA Team Lead	**APPROVED**	28/09/2026
Security Lead	Cybersecurity Auditor	**APPROVED**	28/09/2026
Domain Expert	Legal Consultant	**APPROVED**	28/09/2026
Project Manager	Project Lead	**APPROVED**	28/09/2026

Legal Years - Phase 6: Testing Phase Officially Completed!